

Why label graphs? The FSD50K label co-occurrence analysis, explained

ATGNN contains two components that go beyond a standard audio classifier: the PLG (Patch-Label Graph) block, where learnable label embeddings attach to regions of the spectrogram, and the LLG (Label-Label Graph) block, where a learned matrix lets each label's embedding influence other labels. A natural examiner question is: what is the evidence that modelling relationships between labels is worth doing at all?

This document answers that question with data. The analysis script (`scripts/label_cooccurrence_fsd50k.py`) uses only the official FSD50K ground-truth annotations (`dev.csv` + `eval.csv`: 51,197 clips, 200 classes) and asks two things: (1) do clips genuinely carry multiple labels, and (2) do labels co-occur with strong, systematic structure that a model could learn?

The logic: if labels mostly appeared alone, or co-occurred randomly, a label graph would be pointless - a classifier predicting each class independently would suffice. But if clips usually carry several labels, and certain labels systematically appear together, then a model that learns those relationships has real signal to exploit.

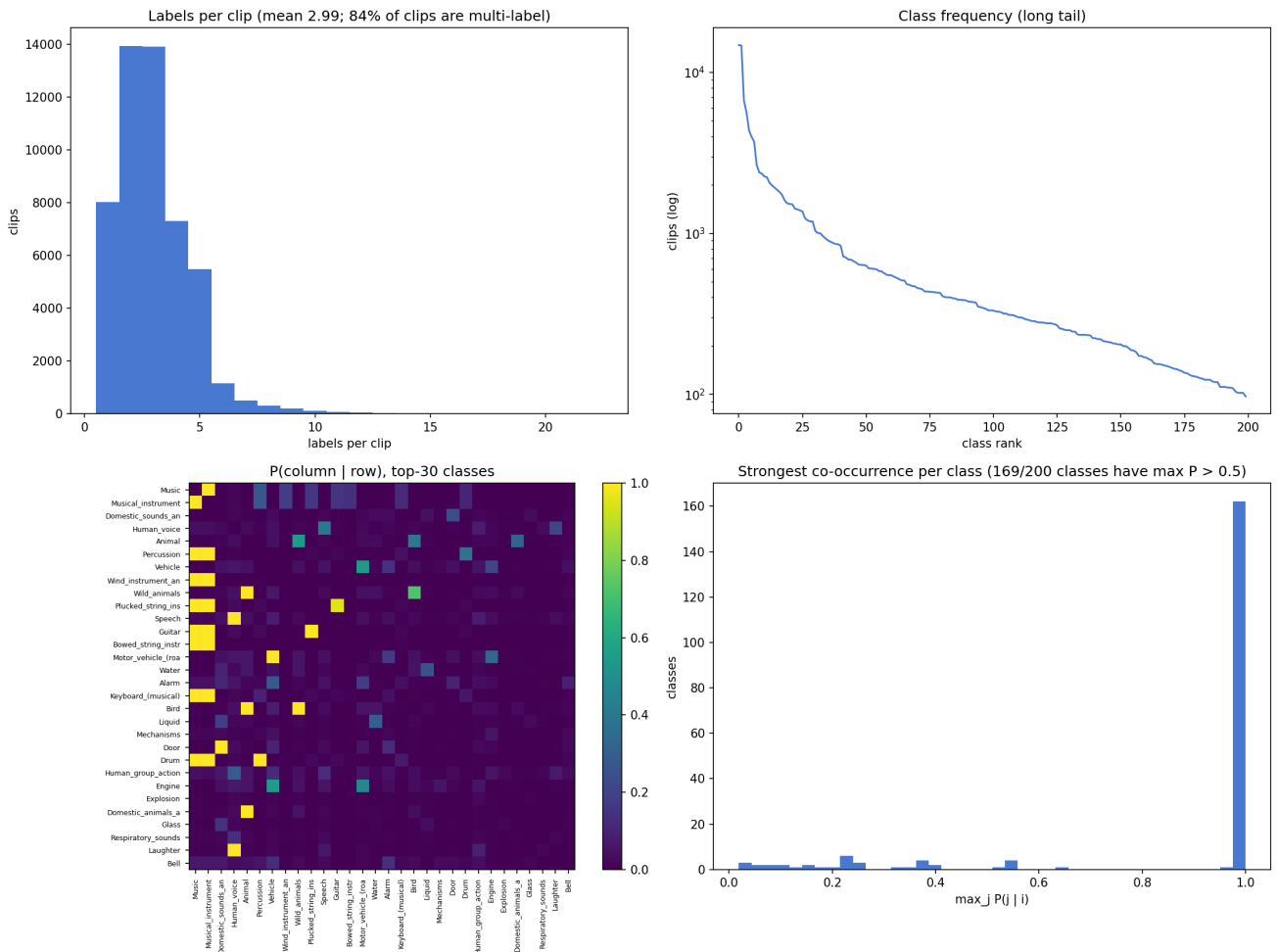
Headline numbers

- 84.3% of clips are multi-label; the mean is 2.99 labels per clip (median 3, max 22).
- 169 of the 200 classes have at least one partner label that co-occurs with them more than half the time - most of those near 100%.
- The strongest pairs: Music + Musical_instrument (14,703 clips, $P \sim 1.0$), Musical_instrument + Percussion (3,977), Animal + Wild_animals (2,392, $P(\text{Wild_animals} \mid \text{Animal}) = 0.55$).

The four-panel figure

The figure below is produced by the analysis script (`results/label_cooccurrence.png`). Each panel is explained on the next page.

FSD50K label statistics (dev + eval ground truth)



Top-left: labels per clip

A histogram of how many labels each clip carries. The bulk of clips have 2-4 labels and 84% have more than one. Takeaway: FSD50K is fundamentally a multi-label dataset, so predicting labels jointly - not independently - matches the data.

Top-right: class frequency (long tail)

Classes ranked from most to least common, on a logarithmic scale. The most common class appears in roughly 15,000 clips, the rarest in about 100 - a steep 'long tail'. Takeaway: rare classes have very little training data of their own, which is exactly the situation where borrowing signal from correlated, more frequent labels helps.

Bottom-left: the co-occurrence heatmap

The core panel. It shows the 30 most frequent classes; the cell at row i , column j is the conditional probability $P(j | i)$: given that a clip contains label i , how likely is it to also contain label j ? Dark purple means the pair almost never co-occurs; bright yellow means that if you see label i , label j is almost certainly present too. For example, the bright cell on the Guitar row under the Music column means virtually every clip tagged Guitar is also tagged Music.

The key visual point: the map is mostly dark with a few very bright cells. Label relationships are sparse and specific, not uniform noise. That specific, learnable pattern is what the LLG block's adjacency matrix can capture.

Bottom-right: how strong is each class's best partner?

For each of the 200 classes we take the highest co-occurrence probability it has with any other class, and plot the distribution. 169 of 200 classes have a partner label that follows them more than half the time - and for most of those the relationship is nearly deterministic (the tall bar at 1.0). Takeaway: almost every class has at least one strongly predictive companion label.

Where the strong pairs come from

Two flavours of co-occurrence show up in results/top_label_pairs.csv:

- Ontology hierarchy. FSD50K labels come from the AudioSet ontology, so an Electric_guitar clip is also tagged Guitar and Music by construction. This gives near-deterministic pairs such as Music + Musical_instrument ($P \sim 1.0$).
- Real-world regularities. Pairs like Animal + Wild_animals or Bird + Wild_animals are not forced by the ontology but reflect how sounds actually co-occur in recordings (normalised PMI 0.80 for Animal + Wild_animals).

Both kinds are learnable structure: if the model is fairly confident about Electric_guitar, the label graph lets it raise its confidence in Music essentially for free - and conversely, implausible label combinations can be suppressed.

How this maps to the ATGNN architecture

- PLG (patch-label): with ~ 3 labels per clip, different labels typically correspond to different regions/times of the spectrogram. Attaching each label embedding to its nearest patch nodes lets co-occurring labels bind to different evidence in the same clip.
- LLG (label-label): the sparse, near-deterministic co-occurrence structure measured here is exactly the kind of latent correlation a learnable label-label adjacency matrix ($L_{\text{hat}} = A L + L$) can encode.

One-sentence summary for the thesis

FSD50K clips carry ~3 labels on average, 84% are multi-label, and nearly every class has a companion label that co-occurs with it more than half the time - so architectures like ATGNN's PLG/LLG blocks, which explicitly model patch-label and label-label relationships, address measurable structure in the data rather than an assumed one.

Reproducing

```
python scripts/label_cooccurrence_fsd50k.py --gt_dir $DATA_DIR/ground_truth --out_dir results
```

Outputs: results/label_cooccurrence.png (the figure), results/label_stats.csv (per-class frequencies), results/top_label_pairs.csv (top-100 pairs with conditional probabilities and normalised PMI). This PDF: python examples/make_label_analysis_pdf.py.